AeroFlux Stage 2: A Scalable, Portable Big-Data Platform for Real-Time Flight-Delay Intelligence

Final Project Report

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Abstract

TODO: abstract goes here (write last).

1 Introduction

Flight delays are among the most costly and disruptive failures in the U.S. National Airspace System (NAS), and they rarely occur in isolation. A single late arrival propagates downstream through the same aircraft's rotation and through shared airport departure and arrival queues, so a local disturbance becomes a network-wide disruption.

Although airlines, airports, and air traffic authorities increasingly use real-time and predictive decision-support systems, these capabilities are often specialized around particular organizations, operational domains, or stages of flight. For example, the FAA's Traffic Flow Management System supports demand-and-capacity management across the National Airspace System, while Terminal Flight Data Manager integrates surface and national traffic-management data to improve airport operations (Federal Aviation Administration, 2020, 2024c). Commercial platforms such as Sabre Mosaic support airline disruption recovery, and Assaia ApronAI uses computer vision and machine learning to monitor aircraft turnarounds and predict departure readiness (Assaia International AG, n.d.; Sabre Corporation, n.d.). Opportunities therefore remain for integrated research platforms that fuse multiple live aviation data streams, model delay propagation across the broader flight network, and support forecasting, simulation, data science, and AI-assisted analysis.

1.1 Problem Statement

TODO: problem statement.

1.2 Contribution

TODO: contribution — the "why should anyone care" argument.

1.3 Scope

The long-term vision for AeroFlux is ambitious and continues to evolve. This project focuses on building a scalable, modular, configurable, portable, and decoupled big-data platform that can ingest, fuse, process, and analyze aviation and weather data in real time, establishing the technical foundation for future AeroFlux capabilities. Flight-delay prediction serves as the primary use case, but the architecture is designed to support additional applications, including congestion analysis, delay-propagation modeling, operational monitoring, decision support, and agentic AI. Stage 1 established a historical machine-learning pipeline and interactive prototype (Wilson et al., 2026); this work extends that with a live data platform, baseline-vs-advanced model comparison, cloud deployment, live-prediction evaluation, and an explainable analyst agent prototype.

1.4 Technical Challenges

Building this platform requires solving several non-trivial problems: (1) ingesting heterogeneous, error-prone streaming sources (FAA SWIM XML, ADS-B, weather) without data loss; (2) *entity resolution* across systems that share no universal key — for example, the aircraft tail number is not published in SWIM flight data and must be recovered from ADS-B, which itself has limited coverage; (3) unstable identifiers, since a flight's `flight_ref` changes on plan amendment while its Globally Unique Flight Identifier (GUFI) remains stable; (4) train/serve skew, because historical labels (Bureau of Transportation Statistics) and live features are measured differently; and (5) delivering all of this at demo quality on a portable, low-cost footprint.

2 Related Work

Machine-learning delay prediction. Prior work frames delay prediction as supervised classification or regression over historical records. The review by Sternberg et al. (2017) describes the major approaches used for flight-delay prediction, while more recent work continues to evaluate machine-learning algorithms and class-imbalance handling (AlBassam & AlShahrani, 2025). These approaches can achieve strong predictive performance on curated historical datasets. However, when used in isolation, they may remain largely retrospective, rely on a limited set of data sources, and model flights as independent observations. These assumptions can restrict their ability to capture the temporal, spatial, and operational dependencies through which delays propagate across the aviation network.

Delay-propagation modeling. A separate literature models how delays ripple through airline schedules. Beatty et al. (1999) introduced an early method for evaluating delay propagation through an airline schedule, Wong & Tsai (2012) applied a survival model, and Kafle & Zou (2016) proposed an analytical-econometric approach. These methods capture important propagation mechanisms but are generally offline and disconnected from real-time streaming prediction. Belcastro et al. (2016) demonstrated scalable data mining for flight-delay prediction, establishing the value of big-data tooling while leaving room for tighter integration with real-time data fusion.

Operational data fusion. The state of the art in operational systems includes NASA's Airspace Technology Demonstration 2 (ATD-2) *Fuser*, which fuses multiple aviation data sources into a unified flight record (National Aeronautics and Space Administration, 2021). Its strength is production-grade fused operational data; its limitation for this project is that it is not designed as a lightweight, open, extensible ML/AI experimentation platform. This project leverages some of the high-level concepts used in ATD-2 to create its own fuser.

Emerging aviation intelligence systems. Recent operational systems show that the field is moving beyond static delay prediction toward real-time prediction, simulation, and decision support. FlightAware Foresight represents production ML-driven predictive flight tracking, including arrival-time, taxi-out, and runway predictions (FlightAware, {n.d.}). NASA's NAS Digital Twin and Project Bluebird extend this direction toward replayable and probabilistic simulation environments for live or historical airspace operations and

AI-agent evaluation (Lauderdale et al., 2024; Pepper et al., 2026). Commercial platforms such as Airspace Intelligence PRESCIENCE, Lufthansa Systems NetLine/Ops++ aiOCC, and Amadeus Flight Operations Control show the market moving toward 4D operating-domain twins, AI-assisted operations control, disruption recovery, and resource-aware decision support (Airspace Intelligence, {n.d.}; Amadeus, {n.d.}; Lufthansa Systems, 2025). Recent prototypes such as FlightSense further combine rotation-chain features, real-time MLOps, and conversational/agentic interfaces for delay prediction (Shelke et al., 2026), while the FAA AI safety roadmap highlights the need for assurance, incremental deployment, and aviation-specific safety methods when adding AI to operational contexts (Federal Aviation Administration, 2024a). Together, these efforts motivate AeroFlux as a lightweight, SWIM-native experimentation platform that connects real-time data fusion, propagation-aware ML, replay, and explainable/agentic analysis rather than treating delay prediction as a standalone model.

Relevance of this approach. AeroFlux Stage 2 bridges these strands. It couples lightweight, Fuser-inspired real-time fusion and entity resolution with propagation-aware feature engineering and an explainable retrieval-augmented agent — integrating capabilities that prior work often treats separately.

3 Data and Fusion

3.1 Data Sources

The platform fuses live and historical aviation and weather sources, plus reference dimensions and a documentation corpus, summarized in Table 1. FAA SWIM provides the authoritative live NAS flight state (Federal Aviation Administration, 2024b); ADS-B supplies the physical airframe identity (ICAO 24-bit address and registration) needed to link an aircraft's successive legs, drawn from community feeds (adsb.lol, 2026; airplanes.live, 2026), whose research value is demonstrated by large-scale networks such as OpenSky (Schäfer et al., 2014); NOAA weather — a leading delay driver — is drawn from live METAR via the Aviation Weather Center (National Oceanic and Atmospheric Administration, 2026) for serving and the NCEI Integrated Surface Database for historical training (National Centers for Environmental Information, 2026); and the BTS On-Time Performance dataset, which includes actual gate times and tail numbers, provides historical training labels (Bureau of Transportation Statistics, 2026). Reference dimensions for airports and airlines (OpenFlights, 2026; OurAirports, 2026) support entity resolution, code normalization, and timezone alignment.

Table 1: AeroFlux Stage 2 data sources, retention policies, and estimated storage requirements. Estimated sizes are project-planning ranges rather than published source totals and depend on geographic scope, ingestion frequency, selected fields, compression, and duplicate handling.

Source	Type	Role	Retention	Estimated retained size
FAA SWIM (TFMS)	Streaming XML	Live flight state (primary)	48 h	Approximately 5–25 GB
ADS-B (airplanes.live / adsb.lol)	Streaming JSON	Airframe and tail resolution	48 h	Approximately 2–10 GB
NOAA weather — METAR (AWC) + NCEI ISD hourly	Polled JSON / text	Weather: METAR for serving, NCEI for training	48 h	Approximately 50–250 MB live
BTS On-Time Performance	Historical CSV / Parquet	Historical training labels	Persistent	Approximately 20–40 GB as CSV or 4–10 GB as Parquet
Aviation documents (FAA, SWIM dictionaries, and related references)	PDF / HTML / text	RAG documentation corpus	Persistent	Approximately 0.5–2 GB, including extracted text and embeddings
Reference dimensions (airports, airlines)	Static CSV	Entity resolution, ICAO/IATA + timezone normalization	Persistent	< 50 MB

3.2 Fusion and Canonical Records

A raw SWIM message (abridged) is shown in Listing 1; the corresponding fused, validated canonical record produced by the ingestion pipeline is shown in Listing 2. Each SWIM document explodes into one record per flight, which the pipeline fuses by GUFI across many messages over time.

TODO: expanded fusion description and tier diagram/figure.

Listing 1 Raw SWIM TFMS flight message (abridged).


```xml
<fdm:fltdMessage acid="AAL2033" airline="AAL" arrArpt="KLGA" depArpt="KCLT"
   flightRef="150976233" msgType="FlightModify" ...>
  <nxcm:flightStatus>PLANNED</nxcm:flightStatus>
  <nxcm:aircraftModel>A320</nxcm:aircraftModel>
  <nxcm:flightTimeData airlineOutTime="2026-07-04T15:39:00Z"
                       airlineInTime="2026-07-04T17:35:00Z"/>
</fdm:fltdMessage>
```


Listing 2 Fused, schema-validated canonical flight-instance record.


```json
{ "flight_instance_id": "KN6770564K", "callsign": "AAL2033",
  "flight_number": "AA2033", "carrier_name": "American Airlines",
  "origin": "KCLT", "destination": "KLGA", "aircraft_type": "A320",
  "scheduled_gate_departure": "2026-07-04T15:39:00Z",
  "estimated_arrival": "2026-07-04T17:41:00Z", "flight_status": "PLANNED",
  "resolution_status": "airline_resolved", "schema_version": "1.0" }
```

4 Methods

4.1 System Architecture

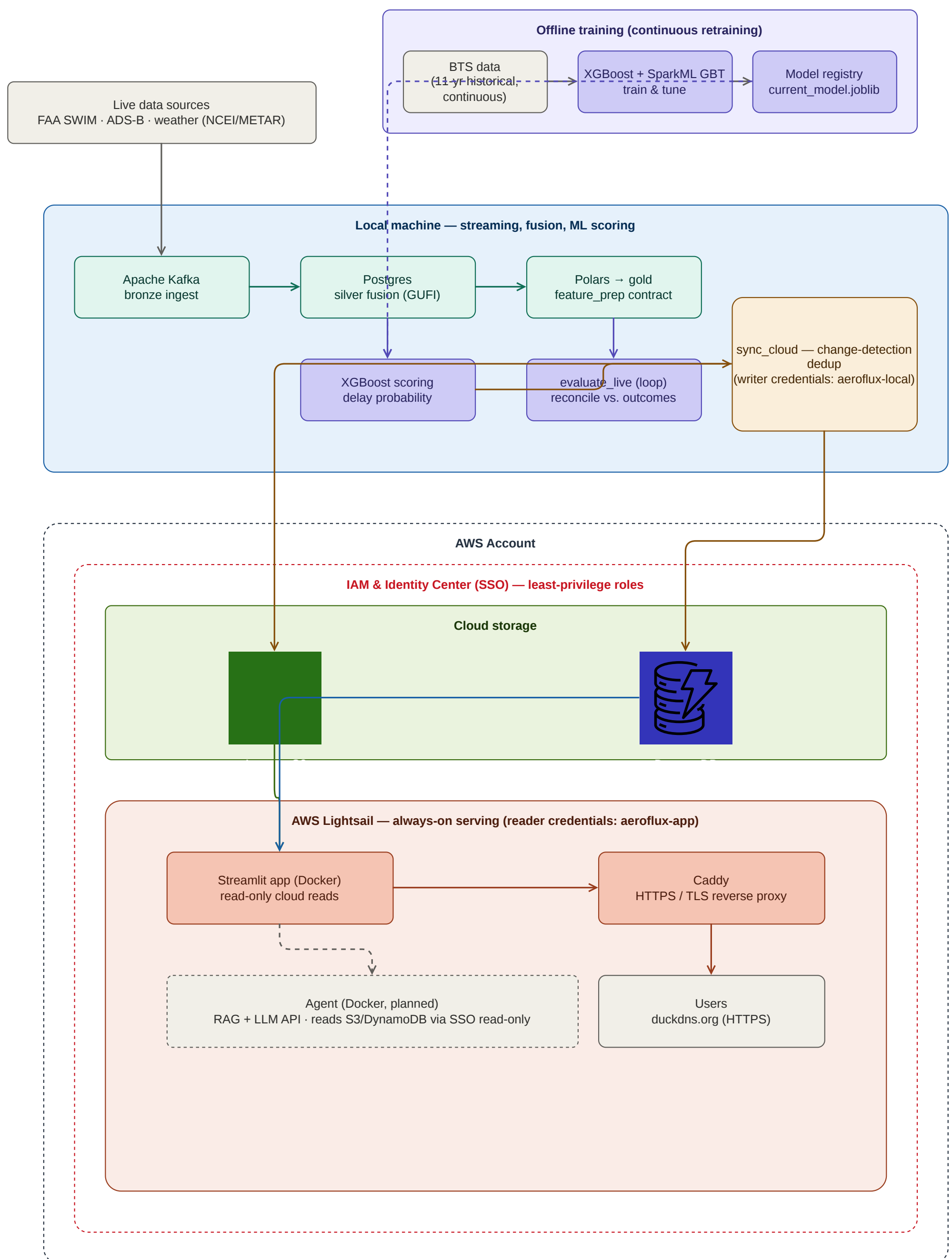

Figure 1: AeroFlux Stage 2 system architecture.

TODO: architecture prose.

4.2 Ingestion and Streaming

TODO: ingestion + streaming.

4.3 Feature Engineering

TODO: feature engineering + the parity argument.

4.4 Modeling

TODO: models.

4.5 Training and Validation Strategy

TODO: train/test strategy, tuning, leakage prevention.

4.6 Live-Prediction Evaluation

TODO: live-eval methodology.

4.7 Agent Prototype

TODO: agent prototype (coordinate with Ryan).

5 Evaluation and Results

We evaluate at two levels, each against an explicit baseline: predictive (model) performance and system performance.

5.1 Predictive Performance

TODO: predictive results — BTS vs. live comparison table + calibration curve.

Table 2: Predictive performance, historical training vs. live reconciled evaluation.

Metric	BTS training (held-out)	Live (reconciled)
ROC-AUC	*TODO*	*TODO*
PR-AUC	*TODO*	*TODO*
F1	*TODO*	*TODO*
Brier	*TODO*	*TODO*
n	*TODO*	*TODO*

5.2 Feature Importance

TODO: feature importance figure + discussion.

5.3 System Performance

TODO: system performance table (baseline vs. achieved).

5.4 Expected vs. Actual Performance Targets

TODO: targets vs. achieved table.

6 Discussion

6.1 Key Concepts and Definitions

TODO: concepts + glossary.

6.2 Assumptions

TODO: assumptions.

6.3 Limitations

TODO: limitations (distribution shift, censoring, coverage, parity gaps).

6.4 What Was Proposed vs. Delivered

TODO: proposed vs. delivered table.

7 Technology Stack

Table 3: Technology stack and how each component was used.

Category	Tool	How it was used
Language	Python 3.11	*TODO*
Streaming	Apache Kafka	*TODO*
Processing	Polars (Spark Structured Streaming as scale path)	*TODO*
Storage (local)	PostgreSQL	*TODO*
Storage (cloud)	Amazon S3, DynamoDB	*TODO*
ML	scikit-learn, XGBoost, SparkML GBT, SHAP	*TODO*
Serving / UI	Streamlit, Plotly	*TODO*

Category	Tool	How it was used
Deployment	Docker, Caddy, GHCR, GitHub Actions, AWS Lightsail	*TODO*
Security	AWS IAM, Identity Center (SSO)	*TODO*
AI orchestration	LangGraph, RAG, LLM API (Bedrock/Anthropic)	*TODO (Ryan)*

8 Conclusion and Future Work

TODO: conclusion + future work.

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